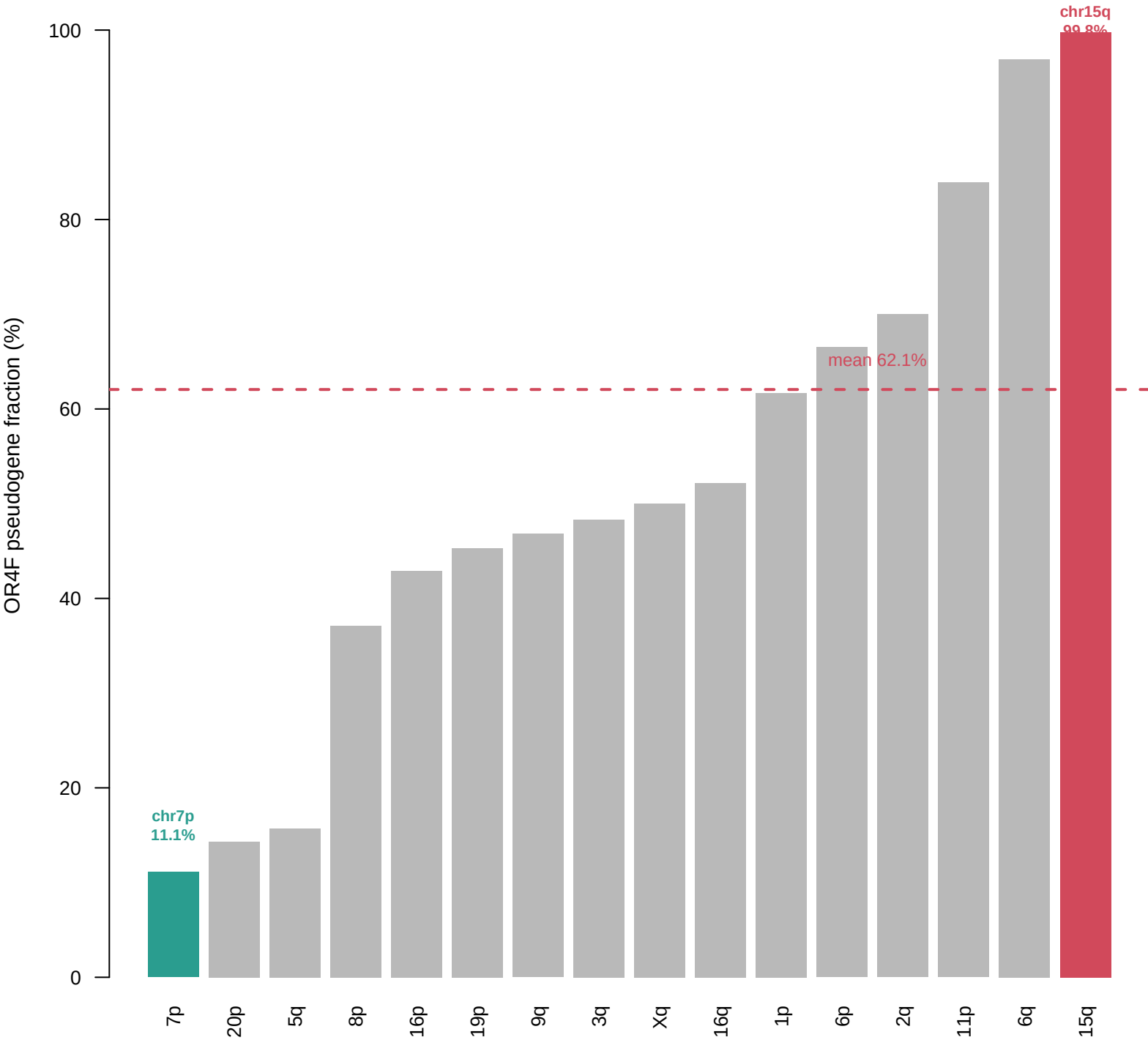


OR4F pseudogenization gradient

5,023 annotations across 16 arms; 3,117 pseudogene / 1,906 coding



Gene–family signal to pair with OR4F

Use the OR4F gradient as the visible plot; keep enrichment wording precise.

OR4F decay gradient

5,023 OR4F annotations across 16 arms; 62.1% pseudogene overall

Use as canonical slide metric; source is HPRCv2 or4f_pseudogene_fraction.csv

OR family community signal

10 OR4F genes in 7 communities; OR4F5 on 14 arms; OR4F8P on 14 arms

Presence pattern, not BH–significant enrichment: C3 OR q=0.071

High–copy coding families

Recovered ORA table: OR4F=72, DUX4/FRG2/FRG2B=54, IL9R/IL9RP=58 copies

Historical Erik copy–aware ORA; parked under paper_prep/_brainstorming, use only as context

D4Z4 / DUX4L community

C1 chr4q/chr10q carries D4Z4; section 9 reports 22 DUX4L pseudogenes specific to C1

Use as linked biological comparator, not as an OR4F substitute

Subtelomeric duplicon backbone

RPL23AP45 10 communities / 21 arms; SEPTIN14P22 9 communities / 22 arms; DDX11L16 9 communities / 20 arms

Canonical community–enrichment qualitative signal; Fisher tests do not survive BH

Fisher note: 116 community–family tests; none BH–significant. Best OR row C3 q=0.071; C7 MTCO q=0.071.